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6618 Advanced Biostatical Methods I
Check in 1 for Final Project
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The dataset I would like to use is from the research facility Scripps Institution of Oceanography. The data was initially published in the primary article "*Climatic and economic fluctuations revealed by decadal ocean soundscapes*" by ZoBell et al. and made publicly available on Dryad as of June 25, 2026. This contains acoustical data for blue and fin whales and potential anthropogenic and environmental sound levels recorded for 15 years, across six sites in Southern California Bight. Each sheet of data, per site, contains approximately 5650 observations.

Variables found in each Excel for the six sites:

(The unit of sound pressure level (SPL) is dB re 1 μPa^2)

* site: site name
* year: Year of SPL measurement
* month: Month of SPL measurement
* day: Day of SPL measurement
* Fin: Fin whale associated daily median sound pressure level.
* Blue: Blue whale-associated daily median sound pressure level.
* Ship: Ship-associated daily median sound pressure level.
* Wind: Wind associated daily median sound pressure level.

Sample of excel sheet:

site	year	month	day	fin	blue	ship	wind
C	2008	2	13	88.89	92.38	87.77	66.69
C	2008	2	14	85.86	92.34	85.55	68.71
C	2008	2	15	85.61	89.47	87.01	63.83
C	2008	2	16	84.69	92.19	88.68	65.56

Each row is the daily median sound pressure levels for fin whales, blue whales, ship noise, and wind at the given site. The primary outcome variables include the sound pressure level of fin and blue whale, which are continuous. The predictor variables will be ship-associated sound pressure level and wind-associated sound pressure level, which are also continuous. Site, year, month, and day are categorical with more than three levels (Ex: Site has six levels and month has 12 levels). There is not an explicit binary variable, but I could do something along the lines of 1/0 for warmer and colder months or 1/0 for presence/no presence of ship (No occurrence in dataset is currently shown as NaN). There may not be total independence of observations as the data does consistent of multiple daily measurements across the same site for years.

Questions I would potentially like to investigate:

1. Is ship-associated SPL associated with fin and blue whale-associated SPL after accounting for wind, site, month, and year?

a. Is ship-associated SPL associated differently with fin whale SPL compared with blue whale SPL?
b. Does the association between ship-associated SPL and whale-associated SPL differ across years?
c. Does the association between ship-associated SPL and whale-associated SPL differ across months?

2. Is wind-associated SPL associated with fin and blue whale-associated SPL after accounting for ship SPL, site, month, and year?
3. Does the association between ship-associated SPL and whale-associated SPL differ between warmer and cooler months?

Citations

Dryad direct link to data:
https://doi.org/10.5061/dryad.nzs7h450t

Primary article from The Journal of the Acoustical Society of America:
ZoBell, V. M., Posdaljian, N., Lenssen, K. L., Wiggins, S. M., Hildebrand, J. A., Baumann-Pickering, S., & Frasier, K. E. (2025). Climatic and economic fluctuations revealed by Decadal Ocean Soundscapes. *The Journal of the Acoustical Society of America*, *157*(6), 4233–4251. https://doi.org/10.1121/10.0036778